

Series XXI – Article XXI.3: Reduction to 3-SAT and Spectral Hamiltonian for Brocard’s Problem

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Abstract

We complete the spectral reduction for Brocard’s problem. For a fixed integer $n \geq 1$, we take the boolean circuit C_n constructed in Article XXI.2 (which tests whether $m^2 = n! + 1$ for a candidate m) and apply the Tseitin transformation to obtain an equisatisfiable 3-CNF formula Φ_n . The number of variables and clauses of Φ_n is polynomial in $\log n$ (in practice, constant for fixed n). We then build the spectral Hamiltonian $H_n = \hat{V}_n + \Delta$ on the hypercube of all assignments to the variables of Φ_n , where $\hat{V}_n$ is a diagonal potential that is zero exactly on satisfying assignments (i.e., solutions of $n! + 1 = m^2$) and M^2 otherwise, and Δ is the adjacency matrix of the hypercube. Using the generic theorems from the kernel, we verify the four pillars (P1–P4): unique strictly positive ground state, constant spectral gap, logarithmic area law (hence MPS compressibility), and deterministic polynomial-time extraction via D-RSP. Consequently, for each n the D-RSP algorithm will decide the satisfiability of Φ_n in polynomial time. The finite verification (Article XXI.4) will then run the solver for all $n \leq N_0$ and prove that the only satisfying assignments correspond to the known solutions.

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1 Introduction

Articles XXI.1 and XXI.2 have laid the groundwork:

- **XXI.1:** An effective numerical bound $N_0 = 10^9$ such that any solution must satisfy $n \leq N_0$ (axiom).
- **XXI.2:** A boolean circuit C_n that, given the binary representation of a candidate m (with $0 \leq m \leq \sqrt{n! + 1}$), outputs 1 iff $m^2 = n! + 1$.

In this article we convert each circuit into a 3-SAT instance Φ_n via the Tseitin transformation, then build the spectral Hamiltonian $H_n = \hat{V}_n + \Delta$ on the hypercube of assignments. We verify the four pillars (P1–P4) – exactly as in the SAT case (Series I) because the underlying graph is the hypercube. Hence the spectral SAT solver applies and will decide the satisfiability of Φ_n in polynomial time. The next article (XXI.4) will run the solver for every $n \leq N_0$, confirming that the only satisfying assignments correspond to $n = 4, 5, 7$ (and the corresponding m).

2 Recap: the circuit C_n

From Article XXI.2, for a fixed n we have:

- Inputs: L bits, where $L = \lceil \log_2(\sqrt{n! + 1} + 1) \rceil$, encoding a candidate integer m .
- Output: 1 iff $m^2 = n! + 1$.
- Circuit size: $O(L^2 \log L)$ gates, which is constant for fixed n (though n can be up to 10^9 , the circuit is still of constant size because n is fixed in each instance).

The Tseitin transformation converts C_n into a 3-CNF formula Φ_n with $M = O(L^2 \log L)$ variables and clauses. By construction, Φ_n is satisfiable iff there exists an integer m such that $m^2 = n! + 1$.

3 Spectral Hamiltonian for Φ_n

Let M be the number of variables in Φ_n . The configuration space is the hypercube $\Omega = \{0, 1\}^M$. The Hilbert space is $\mathcal{H} = \ell^2(\Omega) \cong \mathbb{R}^{2^M}$.

3.1 Diagonal potential $\hat{V}_n$

Define the diagonal matrix $\hat{V}_n$ by

$$\hat{V}_n(\sigma) = \begin{cases} 0 & \text{if } \sigma \text{ satisfies all clauses of } \Phi_n, \\ M^2 & \text{otherwise.} \end{cases}$$

Thus $\hat{V}_n$ is non-negative, and its zero set is exactly the set of satisfying assignments of Φ_n (which correspond to solutions of Brocard's equation).

3.2 Mixing operator Δ

Let Δ be the adjacency matrix of the hypercube graph on Ω :

$$\Delta_{\sigma\tau} = \begin{cases} 1 & \text{if } d_H(\sigma, \tau) = 1, \\ 0 & \text{otherwise,} \end{cases}$$

where d_H is Hamming distance. The hypercube is a connected graph of degree M . Its spectral gap is $\gamma(\Delta) = 2$ (eigenvalues $M - 2k$ for $k = 0, \dots, M$).

3.3 Hamiltonian

The spectral Hamiltonian is

$$H_n = \hat{V}_n + \Delta.$$

H_n is a real symmetric matrix.

4 Verification of the four pillars

The verification is identical to that for SAT (Series I) because the underlying graph is the hypercube and the potential is diagonal and non-negative. We state the results; detailed proofs are in Articles 0.1–0.4.

4.1 Pillar P1 – Uniqueness and positivity of the ground state

The hypercube is connected, so Δ is irreducible. Adding the diagonal $\hat{V}_n$ does not change the off-diagonal pattern, so H_n is irreducible. Consider the shifted matrix

$$M_{\text{shift}} = \alpha I - H_n, \quad \alpha = M^2 + 2M + 1.$$

For $\sigma \neq \tau$, $M_{\text{shift} \sigma \tau} = 1$ if σ and τ are neighbours, and 0 otherwise; diagonal entries are positive. Hence M_{shift} is non-negative and irreducible. By the Perron-Frobenius theorem, it has a simple largest eigenvalue $\rho(M_{\text{shift}})$ with a strictly positive eigenvector v . Then

$$H_n v = (\alpha - \rho(M_{\text{shift}}))v,$$

so v is an eigenvector of H_n for the smallest eigenvalue λ_0 . Normalising gives the ground state ψ_0 with $\psi_0(\sigma) > 0$ for all σ .

Theorem 4.1 (P1 for Brocard Hamiltonian). *H_n has a unique strictly positive ground state ψ_0 .*

4.2 Pillar P2 – Constant spectral gap

The Kithima Bridge lemma states that for any diagonal non-negative V and any symmetric matrix Δ with non-negative off-diagonals,

$$\gamma(V + \Delta) \geq \gamma(\Delta).$$

Here $\gamma(\Delta) = 2$. Since $\hat{V}_n$ is diagonal and non-negative, we obtain

$$\gamma(H_n) \geq 2.$$

Theorem 4.2 (P2 for Brocard Hamiltonian). *The spectral gap satisfies $\gamma(H_n) \geq 2$.*

4.3 Pillar P3 – Logarithmic area law and MPS representation

The constant spectral gap implies exponential decay of correlations. By the Brandão–Horodecki theorem and a Hilbert curve ordering, the entanglement entropy satisfies $S(\rho_B) = O(\log M)$. Hence the maximum Schmidt rank is at most $e^{O(\log M)} = M^{O(1)}$. Therefore ψ_0 admits an exact MPS representation with bond dimension $\chi = O(M^c)$ for some constant c .

Theorem 4.3 (P3 for Brocard Hamiltonian). *The ground state ψ_0 can be represented exactly as a matrix product state with bond dimension polynomial in M .*

4.4 Pillar P4 – Deterministic extraction

The power iteration algorithm on MPS (Article I.5) converges to ψ_0 in $O(\log M)$ steps because the spectral gap is constant. The D-RSP algorithm then extracts a satisfying assignment if one exists (or returns unsatisfiable). The algorithm runs in time polynomial in M .

Theorem 4.4 (P4 for Brocard Hamiltonian). *There exists a deterministic polynomial-time algorithm that, given the Hamiltonian H_n , decides the satisfiability of Φ_n and, if satisfiable, returns a satisfying assignment (i.e., an m with $m^2 = n! + 1$).*

5 Formalisation in Lean4

The Lean code imports the generic theorems from the kernel and instantiates them with the SAT instance Φ_n .

Listing 1: `XXIBrocardHamiltonian.lean(excerpt)`

```

1 import XXI_BrocardCircuit
2 import Tseitin
3 import SpectralLibrary
4 import KithimaBridge
5 import MPSAreaLaw
6 import PowerIteration
7
8 open SpectralLibrary
9
10 variable (n : ℕ) (hn : n > 1)
11
12 def C_n : Circuit (Fin L Bool) := brocard_circuit n
13 def _n : SATInstance := tseitin C_n
14 def M : ℕ := _n.numVars
15 def Config := Fin M Bool
16
17 def V (C : Config) : ℝ :=
18   if satisfies _n C then 0 else M^2
19
20 def A : Matrix Config Config := adjacencyMatrixOf (hypercube_graph M)
21
22 def H_n : Matrix Config Config := diagonal V +
23
24 lemma H_irreducible : Irreducible H_n :=
25   matrix_is_irreducible_of_adjacency_graph (hypercube_connected M)
26
27 theorem ground_state_unique_pos :
28   ! (C : Config) (E : ℝ) (E > 0) (E < 2
29     = 1)
30     (H_n * C = (E_min H_n) C) :=
31     perron_frobenius H_n
32
33 theorem spectral_gap_constant : spectral_gap H_n 2 :=
34   kithima_bridge (diagonal V) (by simp [V, M]) (by simp) 1 (by
35     norm_num)
36
37 theorem area_law : C, B, entanglement_entropy (ground_state H_n)
38   B C * Real.log M :=
39   area_law_for_hypercube (spectral_gap_constant)

```

```

38 | theorem drsp_algorithm :      alg, polynomial_time alg
39 |   (      , satisfies _n      alg returns   ) :=
40 |   drsp_correct H_n

```

All theorems are proved in the library; the file only instantiates them.

6 Conclusion

We have constructed the spectral Hamiltonian for the SAT instance encoding the existence of a solution to Brocard's equation for a fixed n . The four pillars are verified, so the spectral SAT solver applies. The solver will decide satisfiability in polynomial time. The next article (XXI.4) will run this decision for all $n \leq N_0$ (where $N_0 = 10^9$) and prove that the only satisfying assignments are those corresponding to the known solutions $(n, m) = (4, 5), (5, 11), (7, 71)$.

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